

Credit Card Spending Analytics — India

Advanced SQL Portfolio Project | Fathima Suzain

Project Overview

This project analyzes 26,051 real credit card transaction records across major Indian cities to uncover spending patterns by city, card type, expense category, gender, and time period. The analysis was built entirely in SQL (MySQL Workbench), progressing from basic aggregations to advanced window functions and multi-step CTEs, demonstrating end-to-end data cleaning and analytical query design.

Dataset

Source	Kaggle — Credit Card Spending Habits in India
Size	26,051 rows
Columns	id, City, txn_date, card_type, expense_type, Gender, Amount
Card Types	Silver, Gold, Platinum, Signature
Expense Categories	Bills, Food, Fuel, Travel, Entertainment, Grocery

Data Cleaning Steps

- Removed the ", India" suffix from all city name values
- Fixed a byte-order-mark (BOM) corrupted ID column header from the CSV import
- Converted the transaction date column from text to a proper SQL DATE type
- Renamed "Card Type" to "card_type" to remove the space for safe querying

Analysis Structure — 19 Queries

Tier	Queries	Covers
Basic	Q1–Q7	Aggregations & grouping: spend by city, card type, expense category, gender; top-city %
Intermediate	Q8–Q12	Date logic & subqueries: peak spend month overall and per card type, weekend vs week
Advanced	Q13–Q19	Window functions & CTEs: running totals, spend rankings, % contribution per transaction

Key Findings

Fill in with your actual results before sharing — placeholders shown below map to the query that answers them.

- Top-spending city: [CITY] — [X]% of total spend (Query 5)
- Most-used card type by total spend: [CARD TYPE] (Query 2)
- Highest-spend category: [EXPENSE TYPE] — ■[AMOUNT] across [N] transactions (Query 3)
- Gender spend gap: [Male/Female] average transaction ■[AMOUNT] higher (Query 4)
- Peak spending month: [MONTH] (Query 8)
- Weekend vs weekday: [Weekend/Weekday] spending was higher on average (Query 10)

Tools & Skills Demonstrated

MySQL Workbench · Data cleaning (encoding fixes, type conversion, string manipulation) · Aggregate functions (SUM, AVG, COUNT) · Subqueries · Window functions (RANK, ROW_NUMBER, LAG, SUM OVER) · Common Table Expressions (CTEs) · Date/time functions